93. Water is a good solvent of ionic salts because

(a) it has a high boiling point
(b) it has a high dipole moment
(c) it has a high specific heat
(d) it has no colour

Ans. (b) : Water is a good solvent of ionic salts because it has a high dipole moment. Water molecules have a polar arrangement of the oxygen and hydrogen atoms – one side (hydrogen) has a positive electrical charge and the other side (oxygen) had a negative charge. This allows the water molecule to become attracted to many other different types of molecules. Water can become so heavily attracted to a different molecule, like salt NaCl, that it can disrupt the attractive forces that hold the sodium and chloride in the salt molecule together and, thus, dissolve it.

94. If a gas is compressed to half of its original volume at 27°C, to what temperature should it be heated to make it occupy its original volume ?

(a) 327°C (b) 600°C
(c) 54°C (d) 300°C

Ans. (a) : By the Charles's Gas law, the volume occupied by a fixed amount of gas is directly proportional to its absolute temperature, if the pressure remains constant.

i.e. $\boxed{\frac{PV}{T}=k}$

$\therefore \boxed{\frac{P_1V_1}{T_1}=\frac{P_2V_2}{T_2}}$

Where, P_1 is initial pressure, v_1 is initial volume & T_1 is initial temperature.

P_2 = last pressure, V_2 = last volume & T_2 = Last temperature.

From the question p_1, p_2 = constant, $v_1 = 2v$, $v_2 = v$

$T_2 = 27^0C$ or $273 + 27 = 300^0k$, $T_1 = ?$

From the law,

$\frac{v_1}{T_1}=\frac{v_2}{T_2}$ or $\frac{2v}{T_2}=\frac{v}{300}$ or $T_2=\frac{2v\times300}{v}=600k$

Initial temperature = $600 - 273 = 327^0C$

95. The main constituents of pearl are

(a) calcium carbonate and magnesium carbonate
(b) aragonite and conchiolin
(c) ammonium sulphate and sodium carbonate
(d) calcium oxide and ammonium chloride

Ans. (a) : Pearls are formed by calcium carbonate and magnesium carbonate.

BIOLOGY

96. The heart of human embryo starts beating

(a) in the first week of its development
(b) in the third week of its development
(c) in the fourth week of its development
(d) in the sixth week of its development

Ans. (c) : The human heart is one of the first organs to form and function during embryogenesis. By the end of gestational week 3, passive oxygen diffusion becomes insufficient to support metabolism of the developing embryo, and thus the fetal heart becomes vital for oxygen and nutrient distribution. The initiation of the first heart beat via the primitive heart tube begins at gestational day 22, followed by active fetal blood circulation by the end of week 4.

97. If father has blood group A and mother has blood group O, then which one of the following blood group may be found in their son ?

(a) B (b) AB
(c) O (d) B, AB or O

Ans. (c) : Every person has a blood type (O, A, B, or AB) and an Rh factor, either positive or negative. The blood type and the Rh factor simply mean that a person's blood has certain specific features. The blood type is found as proteins on red blood cells and in body fluids. The Rh factor is a protein that is found on the covering of the red blood cells. If the Rh factor protein is on the cells, the person is Rh-positive. If there is not Rh factor protein, the person is Rh-negative.

ABO blood explained in following chart-

	Father's Blood type				
	A	B	A B	O	
Mother's Blood type	A or O	A, B, A B, or O	A, B, or A B	A or O	Child's Blood type must Be
	A, B, A B or O	B or O	A, B, or A B	B or O	
	A, B,	A, B,	A, B,	A or	
	or A B	or A B	or A B	B	
	A or O	B or O	A or B	O	

98. What is the correct sequence of the following in a heart attack ?

I. Narrowing of the inner orifice of the vessel
II. 'Plaque' from fibrous tissue and high cholesterol
III. Inadequate supply of blood and oxygen
IV. Clots of blood carried into the coronary arteries.

Choose the answer from the codes given below :

Codes:

(a) I, II, III, IV (b) II, IV, I, III
(c) II, III, I, IV (d) IV, II, I, III

Ans. (b) : Most heart attacks are the result of coronary artery disease, also known as atherosclerosis or "hardening of the arteries", a condition that clogs coronary arteries with fatty, calcified plaques over time. The typical trigger for a heart attack is often a blood clot that blocks the flow of blood through a coronary artery.

While the step-by-step process leading to a heart attack is not fully understood, major risk factors for coronary artery disease are well-known. You can control some of them including high blood pressure, high cholesterol obesity, smoking and a sedentary lifestyle.

99. Even though an animal is fed with carbohydrate rich diet, its blood sugar concentration tends to remain constant. This is on account of the fact that in the case of an animal

(a) hormones of pituitary glands control metabolic process
(b) hormones of pancreas cause such a condition
(c) blood sugar is readily absorbed by liver
(d) glucose undergoes autolysis

Ans. (b) : Even though an animal is fed with carbohydrate-rich diet, its blood sugar concentration tends to remain constant. This is on account of the fact that in the case of an animal hormones of pancreas cause such a condition.

100. Which of the following are characteristics shown by a breastfed baby when compared to a bottle fed baby ?

I. It is less obese.
II. It shows greater capacity, to resist diseases.
III. It gets more vitamins as proteins.
IV. Its growth in height is abnormal.

Select the correct answer from the codes given below.

Codes:

(a) I, II and III (b) I, II and IV
(c) I, III and IV (d) II, III and IV

Ans. (a) : Breastfed infants is less obese, it shows greater capacity, to resist diseases & it gets more vitamins and proteins when compared to a bottle-fed infants. Breastfed infants have less difficulty with digestion than do bottle-fed infants-so that breastfed babies have fewer bouts of diarrhea or constipation.

101. In the balanced dietaries for a day of an adult working man, weights of carbohydrate food and total protein food (both superior and inferior) should be in order of

(a) 600 gm and 420 gm respectively
(b) 600 gm and 600 gm respectively
(c) 420 gm and 600 gm respectively
(d) 420 gm and 420 gm respectively

Ans. (a) : In the balanced dietaries for a day of an adult working man, weights of carbohydrate food and total protein food should be in order of 600 gm and 420 gm respectively. Proteins are often called the body's building blocks. They are used to build and repair tissues. They help you fight infection.

Carbohydrates are the body's main source of energy. The fruit, vegetables, dairy and grain food groups all contain carbohydrates.

Fat give you energy, and they help the body absorb certain vitamins. Essential fatty acids help the body function, but they aren't made by your body-you have to consume them.

102. Which one of the following animals stores water in the intestine ?

(a) Moloch (b) Camel
(c) Zebra (d) Uromastix

Ans. (b) : Camels are able to survive for long periods of time without either food or water as they store fat in their hump which can be used to give the camel energy when resources are scarce. Camels are an animal with multiple stomachs, having three different stomachs that specialize in absorbing as many nutrients as possible from their harsh environment.

103. Which one of the following is caused by the expression of a recessive gene present on sex chromosome ?